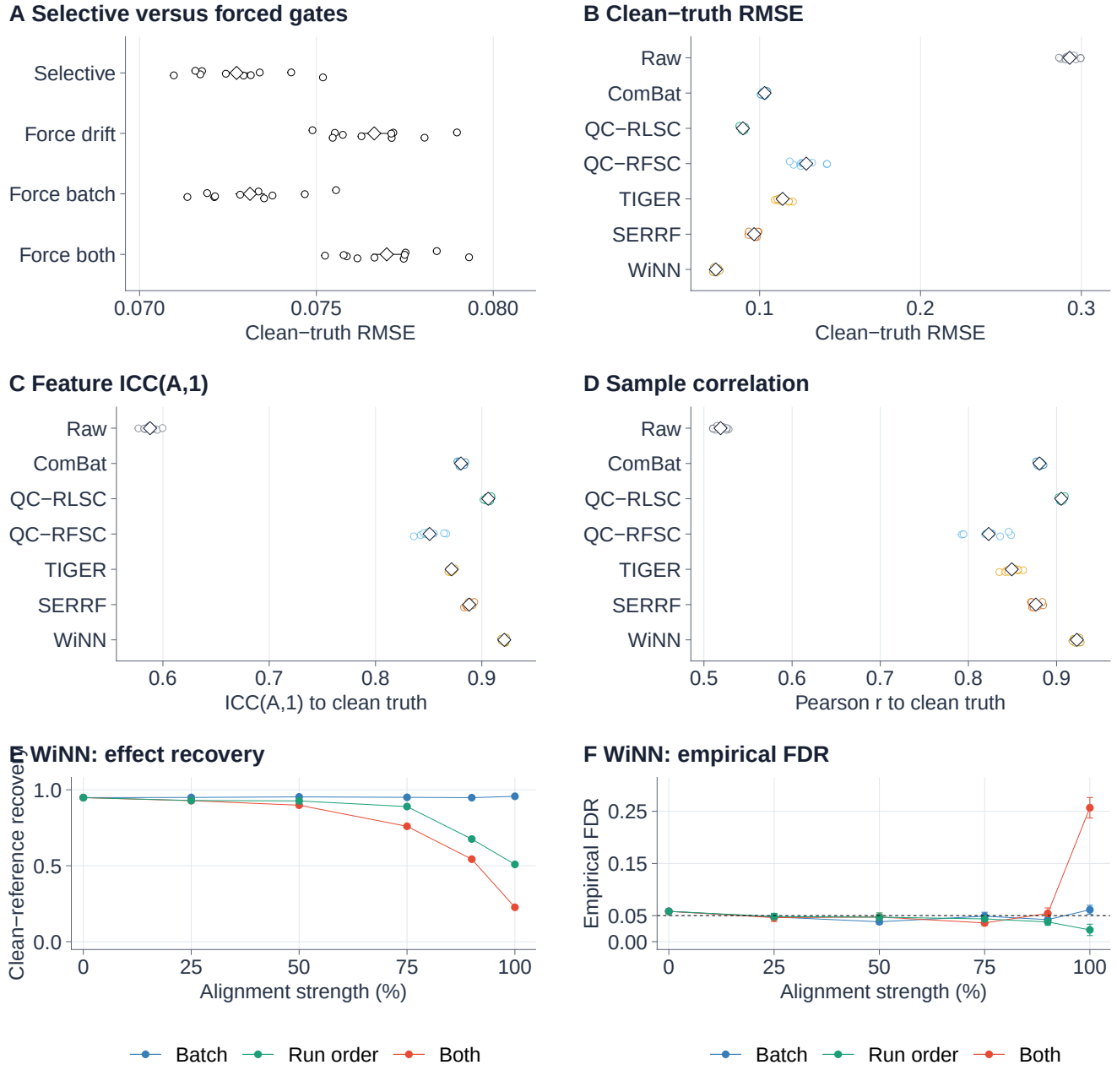

A Selective versus forced gates
Selective
Force drift
Force batch
Force both
0.070
0.075
0.080
Clean−truth RMSE
B Clean−truth RMSE
Raw
ComBat
QC−RLSC
QC−RFSC
TIGER
SERRF
WiNN
0.1
0.2
0.3
Clean−truth RMSE
C Feature ICC(A,1)
Raw
ComBat
QC−RLSC
QC−RFSC
TIGER
SERRF
WiNN
0.6
0.7
0.8
0.9
ICC(A,1) to clean truth
D Sample correlation
Raw
ComBat
QC−RLSC
QC−RFSC
TIGER
SERRF
WiNN
0.5
0.6
0.7
0.8
0.9
Pearson r to clean truth
E WiNN: effect recovery
Clean−reference recovery
1.0
0.5
0.0
0
25
50
75
100
Alignment strength (%)
F WiNN: empirical FDR
Empirical FDR
0.25
0.15
0.05
0.00
0
25
50
75
100
Alignment strength (%)
Batch
Run order
Both
Batch
Run order
Both